TRICKS
& PRO MOVES GRIMOIRE

LINUX COMMAND GRIMOIRE | VOL. 4 | ANIME CHEAT-SHEET EDITION



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36 CHAPTERS * 613 ENTRIES * 6 CODE SCROLLS

01 SHELL SPEED KEYS

Fingers of lightning

Ctrl+W
delete the word before the cursor.

Ctrl+U
cut everything before the cursor.

Ctrl+K
cut everything after the cursor.

Ctrl+Y
paste what you last cut.

Alt+B / Alt+F
move back / forward one word.

Alt+.
insert the last argument of the previous command.

Ctrl+X Ctrl+E
edit the current command in your editor.

^old^new
re-run last command, replacing old with new.

!grep
re-run the last command starting with grep.

!?text?
re-run the last command containing text.

!*
all arguments of the previous command.

fc
edit the last command in an editor, run on exit.

Ctrl+S / Ctrl+Q
freeze / unfreeze terminal output.

Ctrl+D
send EOF: end input or log out.

Ctrl+_
undo the last edit on the line.

history -d 123
delete one history entry (e.g. a typed secret).

(space)command
leading space hides it from history*.

SENPAI SAYS
Learn one shortcut a day. In a month you will fly.

02 BASH MAGIC

Expansion and substitution

mkdir -p proj/{src,docs,tests}
create several folders at once.

cp file{,.bak}
quick backup: expands to cp file file.bak.

echo {1..10}
number ranges (also {a..e} for letters).

ls *.{jpg,png}
match several extensions.

shopt -s globstar; ls **/*.log
** matches files in all subfolders.

${f%.txt}
strip a suffix from a variable.

${f##*/}
strip the longest prefix up to /, like basename.

${f/old/new}
replace the first match in a variable.

${f^^} ${f,,}
uppercase / lowercase (bash 4+).

${#f}
length of a string.

${f:0:5}
substring: 5 chars from position 0.

diff <(sort a) <(sort b)
process substitution: output as a file.

echo $((5 * 3))
integer arithmetic.

arr=(a b c); echo ${arr[1]}
arrays (${arr[@]} gives all items).

declare -A m; m[k]=v
associative array (key/value map).

cat <<EOF > file
here-doc: write many lines until EOF.

grep x <<< "text"
here-string: feed a string as input.

echo $RANDOM $SECONDS
random number / seconds since shell start.

SENPAI SAYS
Brace expansion runs before the command, so preview it with echo first.

03 BASHRC POWER SETUP

Your personal spellbook

source ~/.bashrc
apply changes without logging out.

alias
list your current aliases.

echo $SHELL
which login shell are you using?

export HISTTIMEFORMAT="%F %T "
timestamps in history output.

shopt -s histappend
append to history, don't overwrite it.

shopt -s cdspell
auto-fix small typos in cd.

PS1='\u@\h:\w\$ '
custom prompt: user@host:path.

set -o vi
vi-style line editing (set -o emacs reverts).

complete -p
list tab-completion rules.

export PATH="$HOME/bin:$PATH"
add your own scripts folder to PATH.

SPELL SCROLL | ~/.bashrc

```
alias ll='ls -lah --color=auto'
alias ..='cd ..'
alias k=kubectl
alias gs='git status -sb'
alias dc='docker compose'
export EDITOR=vim
export HISTSIZE=50000
export HISTCONTROL=ignoredups:erasedups
shopt -s histappend
source <(kubectl completion bash)
complete -o default -F __start_kubectl k
```

SENPAI SAYS
Put these in ~/.bashrc, then run source ~/.bashrc to apply them.

04 TMUX PRO MOVES

Never lose a session

- `tmux new -As main` — attach to 'main' or create it.
- `tmux ls` — list sessions.
- `Ctrl+b d` — detach (session keeps running).
- `Ctrl+b c` — new window.
- `Ctrl+b n / p` — next / previous window.
- `Ctrl+b ,` — rename the window.
- `Ctrl+b %` — split pane left / right.
- `Ctrl+b "` — split pane top / bottom.
- `Ctrl+b arrow` — move between panes.
- `Ctrl+b z` — zoom a pane; again to restore.
- `Ctrl+b [` — scroll mode (q to exit).
- `Ctrl+b x` — close the current pane.
- `tmux capture-pane -p` — print pane contents to stdout.
- `tmux kill-session -t main` — delete a session.
- `screen -S name` — GNU screen: new named session.
- `screen -r name` — reattach to a screen session.

SENPAI SAYS
tmux keeps work alive when SSH drops. Use it on every remote server.

05 MODERN CLI UPGRADES

New tools, better defaults

- `bat file` — cat with syntax colors and line numbers. MODERN
- `eza -la --git` — modern ls with git status. MODERN
- `fd pattern` — fast, friendly find. MODERN
- `rg "text"` — ripgrep: very fast recursive search. MODERN
- `fzf` — fuzzy finder for files and history. MODERN
- `z projects` — zoxide: jump to frequent folders. MODERN
- `btop` — beautiful resource monitor. MODERN
- `duf` — readable df replacement. MODERN
- `dust` — visual du replacement. MODERN
- `delta` — nicer git diff pager. MODERN
- `http GET url` — HTTPie: friendly HTTP client. MODERN
- `hyperfine 'cmd'` — benchmark a command. MODERN
- `ls *.py | entr pytest` — re-run tests when files change. MODERN
- `tldr cmd` — community cheat sheets. MODERN
- `lnav /var/log/syslog` — smart log viewer. MODERN
- `glances` — all-in-one system monitor. MODERN
- `fastfetch` — system info with a logo. MODERN

SENPAI SAYS
On Debian/Ubuntu some are named batcat or fdfind. Install via apt, brew or cargo.

06 OPS HELPER TOOLS

The DevOps toolbelt

- `k9s` — terminal UI for Kubernetes. MODERN
- `stern app` — tail logs from many pods at once. MODERN
- `kubectx name` — switch cluster context fast. MODERN
- `kubens name` — switch namespace fast. MODERN
- `lazydocker` — terminal UI for Docker. MODERN
- `ctop` — top-like view of containers. MODERN

`lazygit` MODERN
terminal UI for Git.

`dive image` MODERN
explore Docker image layers.

`trivy image img` MODERN
scan an image for vulnerabilities.

`hadolint Dockerfile` MODERN
lint a Dockerfile.

`tflint` MODERN
lint Terraform code.

`checkov -d .` MODERN
scan IaC for misconfigurations.

`kubeconform f.yaml` MODERN
validate Kubernetes manifests offline.

`pre-commit install` MODERN
run hooks on every commit.

`direnv allow` MODERN
auto-load env vars per folder.

`mise use node@20` MODERN
manage tool versions per project.

`gitleaks detect` MODERN
find secrets committed to git.

SENPAI SAYS
Lint, scan and visualise before you deploy. Cheaper than fixing it in production.

07 TEXT & DATA WRANGLING
Alchemy for logs and JSON

`awk '!seen[$0]++' file`
remove duplicates, keep original order.

`awk '{a[$1]++} END{for(k in a) print k,a[k]}' f`
count lines by first column.

`awk 'length>80' file`
lines longer than 80 characters.

`sed -n '/start/,/end/p' file`
print a block between two patterns.

`sed -i.bak 's/a/b/' file`
in-place edit with a .bak backup.

`sort file | uniq -d`
show only duplicated lines.

`join a.txt b.txt`
join files on a common first field.

`paste -sd, file`
join all lines into one comma list.

`tr '\n' ' ' < file`
turn newlines into spaces.

`shuf -n 3 file`
pick 3 random lines.

`seq 1 5`
print numbers 1 to 5.

`printf "%05d\n" 42`
formatted output: 00042.

`numfmt --to=iec 1048576`
human-readable size: 1.0M.

`jq '.[] | select(.status=="failed")'`
filter JSON array items.

`jq -r '.items[] | [.name,.status] | @tsv'`
JSON to tab-separated rows.

`yq -i '.spec.replicas = 3' f.yaml`
edit YAML in place.

`column -t -s, data.csv`
align CSV columns for reading.

`cut -d, -f2,4 data.csv`
pick CSV columns 2 and 4.

`dos2unix file`
fix Windows line endings (CRLF).

`iconv -f latin1 -t utf-8 f`
convert text encoding.

SENPAI SAYS
awk, sed and jq together can replace many small scripts.

08 FIND & FILE TRICKS
Treasure hunting

`find . -type f -printf '%s %p\n' | sort -rn | head`
the 10 biggest files.

`find . -type f -mmin -10`
files changed in the last 10 minutes.

`find . -newer ref.txt`
files newer than a reference file.

`find . -empty`
empty files and directories.

`find . -user root -type f`
files owned by root.

`find . -iname "*.JPG"`
case-insensitive name search.

`find . -maxdepth 1 -type d`
only one level deep.

`find . -name "*.py" -print0 | xargs -0 wc -l`
safe with spaces in names.

`stat -c '%a %U %n' file`
octal perms, owner and name.

`ls -ltr`
oldest first, newest last.

`ls -S`
sort by size, largest first.

`ls -d */`
list only directories.

`tree -L 2 -d`
directories only, 2 levels deep.

`du -sh * | sort -h`
sizes, smallest to largest.

`install -m 755 -D f /usr/local/bin/f`
copy, set mode, create dirs.

`mktemp -d`
create a safe temporary directory.

`touch -d "2 days ago" file`
set a custom timestamp.

`cp --parents a/b/c.txt dest/`
copy keeping the folder path.

`rename 's/ /_/g' *`
replace spaces in names (perl rename).

`inotifywait -m dir`
watch a folder for file events.

SENPAI SAYS
Use -print0 with xargs -0 so filenames with spaces don't break your pipeline.

09 TIME & DATE FOR OPS
Master of the clock

`date +%F_%H-%M-%S`
filename timestamp: 2026-11-02_14-05-09.

`date -u +%FT%TZ`
ISO 8601 time in UTC.

`date +%s`
Unix epoch seconds.

`date -d @1700000000`
convert epoch to readable time.

`date -d "next friday" +%F`
natural-language dates.

`date -d "+7 days" +%F`
the date one week from now.

`TZ=Asia/Kolkata date`
show time in another timezone.

`timedatectl list-timezones | grep Asia`
find a timezone name.

`sudo timedatectl set-timezone Asia/Kolkata`
change the system timezone.

`hwclock --show`
hardware clock time.

`chronyc tracking`
NTP sync accuracy (chrony).

`timedatectl timesync-status`
systemd-timesyncd status.

`uptime -s`
exact time the system booted.

`stat -c %y file`
last modification time of a file.

`cal 2026`
calendar for a whole year.

`sleep 90`
wait 90 seconds (also 5m, 2h).

SENPAI SAYS
Store and compare times in UTC. Convert to local time only for humans.

10 HIDDEN GEMS & FUN
Secret scrolls

`watch -d 'cmd'`
highlight what changed between runs.

`pv bigfile | gzip > big.gz`
progress bar inside a pipe.

`yes | cmd`
automatically answer 'y'.

`tee -a log.txt`
append to a file and still show output.

`bc -l <<< "2^10"`
command-line calculator.

`factor 360`
prime factors of a number.

`rev`
reverse each line of text.

`sort f | sponge f`
moreutils: safely rewrite the same file.

`cmd | ts`
moreutils: timestamp every output line.

`figlet Hello`
big ASCII-art text.

`cowsay "moo"`
a cow says your message.

`fortune | cowsay`
random quote from a cow.

`sl`
steam locomotive for when you mistype ls.

`cmatrix`
Matrix-style falling characters.

`notify-send Done`
desktop notification when a task ends.

`lolcat`
rainbow-colour any text stream.

SENPAI SAYS
Add echo -e '\a' after a long job and your terminal will beep when it's done.

11 SSH MAGIC
Teleportation spells

`ssh -J bastion user@target`
jump through a bastion host.

`ssh -N -L 5432:db:5432 bastion`
tunnel a remote DB port to localhost.

`ssh -D 1080 host`
SOCKS proxy through a server.

`ssh -R 9000:localhost:3000 host`
expose a local port on the remote.

`ssh -t host 'sudo cmd'`
run a remote command with a TTY.

`ssh host 'cmd' > local.txt`
save remote output locally.

`sshfs user@host:/dir /mnt/x`
mount a remote folder locally.

`ssh-keygen -lf key.pub`
show a key's fingerprint.

`ssh-keygen -R host`
remove a stale known_hosts entry.

`ssh-add ~/.ssh/id_ed25519`
load a key into the agent.

`ssh -o ConnectTimeout=5 host`
fail fast when unreachable.

`ssh -q host exit; echo $?`
quick reachability test.

`rsync -avzP -e ssh src host:/dst`
resumable copy with progress.

`tar czf - dir | ssh host 'tar xzf - -C /dst'`
stream a folder, no temp file.

`ssh -O exit host`
close a multiplexed connection.

SPELL SCROLL | ~/.ssh/config

```
Host bastion
  HostName 203.0.113.10
  User ubuntu
  IdentityFile ~/.ssh/id_ed25519

Host app-*
  ProxyJump bastion
  User ubuntu

Host *
  ServerAliveInterval 30
  ControlMaster auto
  ControlPath ~/.ssh/cm-%r@%h:%p
  ControlPersist 10m
```

SENPAI SAYS
Put repeated options in ~/.ssh/config. Then ssh app-1 just works.

12 NETWORK QUICK TRICKS
Instant gateways

`python3 -m http.server 8000`
instant web server for this folder.

`ss -tan | awk 'NR>1{print $1}' | sort | uniq -c`
count TCP connections by state.

`ss -o state established '( dport = :443 )'`
established HTTPS connections.

`ip -4 -o addr show`
one line per IPv4 address.

`ip -j addr | jq '.[].ifname'`
IP info as JSON.

`curl -sI url | head -1`
just the HTTP status line.

`curl --retry 3 --max-time 10 url`
retry and set a timeout.

`curl -L -o file url`
follow redirects, save to file.

`curl -u user:pass url`
basic authentication.

`curl -k https://host` DANGER
skip TLS checks (testing only!).

`wget -c url`
resume an interrupted download.

`wget -r -np -k url`
mirror a site for offline reading.

`iperf3 -s / iperf3 -c host`
measure bandwidth between two hosts.

`nmap -sn 192.168.1.0/24`
find live hosts (ping sweep).

`nmap -sV -p 22,80,443 host`
detect service versions.

`tcpdump -A -nn port 80`
show HTTP payload as text.

`nc -zv host 20-25`
scan a small port range.

`getent hosts name`
resolve the way the system does.

`mtr -T -P 443 host`
TCP-mode path test to port 443.

SENPAI SAYS
Only scan or probe systems you own or have written permission to test.

13 PROCESS & RESOURCE PRO
Taming the beasts

`ps -p PID -o etime,cmd`
how long has a process been running?

`ps -eo pid,ppid,%cpu,%mem,cmd --sort=-%mem | head`
custom columns, sorted.

`pgrep -af name`
find processes with full command lines.

`pkill -f "python app.py"`
kill by matching the full command.

`kill -0 PID`
check a PID exists (sends no signal).

`wait`
in scripts: wait for background jobs.

`disown -h %1`
detach a job so it survives logout.

`setsid cmd`
run a command in a new session.

`systemd-run --scope -p MemoryMax=500M cmd`
run with a memory cap.

`systemd-run --on-active=30s cmd`
run a command once, 30 seconds from now.

`ionice -c3 cmd`
run with idle IO priority.

`cpulimit -p PID -l 50`
cap a process at 50% CPU.

`prlimit --pid PID --nofile=4096`
raise open-file limit of a running process.

`stdbuf -oL cmd`
line-buffer output so pipes show it live.

`flock -n /tmp/x.lock cmd`
run only one copy at a time.

`ulimit -a`
show all limits for this shell.

`cat /proc/PID/limits`
limits of a running process.

`lsof -p PID | wc -l`
how many files a process has open.

`timeout -s KILL 30 cmd`
hard-kill after 30 seconds.

`nohup cmd > out.log 2>&1 &`
background job with logging.

SENPAI SAYS
systemd-run gives you cgroup limits without writing a unit file.

14 DISK & FILESYSTEM PRO
Vault engineering

`lsblk -f`
devices with filesystem type, label and UUID.

`df -h --total`
add a grand total line.

`df -hx tmpfs -x devtmpfs`
hide virtual filesystems.

`du -sh .[!.]* *`
include hidden folders in totals.

`findmnt -t ext4,xfs`
show only real disk mounts.

`mount | column -t`
tidy mount list.

`hdparm -tT /dev/sda`
quick disk read speed test.

`dd if=/dev/zero of=t bs=1M count=512 oflag=direct` DANGER
write speed test (delete t after).

`fio --name=t --rw=randread --size=256M`
serious IO benchmark.

`sudo fstrim -av`
trim free blocks on SSDs.

`truncate -s 1G sparse.img`
create a 1 GB sparse file instantly.

`mount -o loop disk.img /mnt`
mount an image file.

`mount -t tmpfs -o size=1G tmpfs /mnt/ram`
RAM-backed fast scratch space.

`rsync -a --link-dest=../prev src/ new/`
space-saving incremental snapshots.

`cp --reflink=auto big copy`
instant copy on Btrfs / XFS.

`stat -f /`
filesystem stats: blocks, inodes, type.

`e2label /dev/sdX1 DATA`
set an ext4 label.

`blkid -s UUID -o value /dev/sdX1`
print just the UUID.

SENPAI SAYS
Use blkid to get UUIDs for /etc/fstab. Device names like sdb can change.

15 SPECIAL PERMS & ACLS
Advanced barriers

`chmod 2775 dir`
setgid: new files inherit the dir's group.

`chmod +t dir`
sticky bit: only owners delete their files.

`chmod u+s file` DANGER
setuid: run as file owner (rarely!).

`chmod -R g+rwX dir`
X adds execute only to dirs and programs.

`umask 027`
new files: owner rw, group r, others none.

`setfacl -d -m g:team:rwx dir`
default ACL for future files.

`setfacl -R -m u:alice:rX dir`
recursive ACL for one user.

`setfacl -b file`
remove all ACLs.

`getfacl -R dir`
show ACLs recursively.

`stat -c '%A %U:%G %n' *`
readable perms, owner and group.

`find / -perm /6000 -type f 2>/dev/null`
files with setuid or setgid.

`sudo -l`
what may I run with sudo?

`sudo -u user cmd`
run a command as another user.

`visudo -c`
check sudoers syntax.

`newgrp group`
switch active group in this shell.

`id -nG`
list your group names.

`chown --reference=a b`
copy owner and group from file a.

`getcap -r / 2>/dev/null`
files with Linux capabilities.

SPELL SCROLL | /etc/sudoers.d/deploy

```
# edit with: sudo visudo -f /etc/sudoers.d/deploy
deploy ALL=(root) NOPASSWD: /usr/bin/systemctl restart app
%devs  ALL=(ALL) ALL
```

SENPAI SAYS
Shared team folder recipe: chmod 2775 plus a default ACL. Files inherit both.

16 KERNEL & /PROC PEEK

Under the hood

`uname -r`
running kernel version.

`cat /proc/cmdline`
kernel boot parameters.

`lsmod | grep name`
is a kernel module loaded?

`modinfo name`
details about a module.

`sudo modprobe name`
load a module and its dependencies.

`sudo modprobe -r name`
unload a module.

`sysctl -a | grep keyword`
search kernel parameters.

`echo "vm.swappiness=10" | sudo tee /etc/sysctl.d/99-x.conf`
make a sysctl change persistent.

`sudo sysctl --system`
reload all sysctl config files.

`sync; echo 3 | sudo tee /proc/sys/vm/drop_caches` DANGER
drop page cache (testing only).

`dmesg --level=err,warn -T`
only errors and warnings, with times.

`getconf _NPROCESSORS_ONLN`
number of online CPUs.

`tr '\0' ' ' < /proc/PID/cmdline`
full command line of a process.

`tr '\0' '\n' < /proc/PID/environ`
environment of a process (root).

`cat /proc/pressure/cpu`
CPU pressure stall information.

`ldd /bin/ls`
shared libraries a binary needs.

`ldconfig -p | grep libname`
is a shared library installed?

`getent passwd user`
look up a user via NSS (LDAP too).

`ls /sys/class/net`
network interface names.

SENPAI SAYS
/proc and /sys are windows into the kernel. Nothing there is stored on disk.

17 SYSTEMD & JOURNAL PRO

Summoner's secrets

`systemd-analyze security app.service`
hardening score for a service.

`systemd-cgls`
control group tree of services.

`systemd-cgtop`
top for control groups.

`systemctl list-dependencies --reverse x`
what depends on this unit?

`systemctl status app --no-pager -l`
full status without paging.

`systemctl kill -s HUP app`
send a signal through systemd.

`journalctl -o json-pretty -n 1`
one log entry with all fields.

`journalctl --since "10 min ago" -p warning`
recent warnings and errors.

`journalctl -u a -u b`
interleave logs from two units.

`journalctl --grep="timeout"`
search the journal by pattern.

`journalctl -F _SYSTEMD_UNIT`
list every unit that has logged.

`journalctl _PID=1234`
logs from one PID.

`sudo hostnamectl set-hostname web01`
set the hostname.

`systemctl list-unit-files --state=enabled`
everything set to start at boot.

`sudo systemctl isolate rescue.target` DANGER
drop to single-user rescue mode.

`loginctl show-session $XDG_SESSION_ID`
details about your session.

`systemctl --user status`
manage per-user services.

SENPAI SAYS
systemd-analyze security gives a hardening score. Aim to lower the exposure.

18 SECURITY QUICK AUDITS

Guardian's checklist

`sudo ss -ltnp`
listening TCP ports with owning process.

`sudo lsof -i -P -n | grep LISTEN`
same idea, using lsof.

`grep 'Failed password' /var/log/auth.log | awk '{print $(NF-3)}' | sort | uniq -c | sort -rn | head`
top IPs failing SSH login.

`sudo awk -F: '($2==""){print $1}' /etc/shadow`
accounts with an empty password.

`sudo find / -nouser -o -nogroup 2>/dev/null`
files with no valid owner.

`getent passwd | awk -F: '$3>=1000{print $1}'`
list regular user accounts.

`sudo sshd -T | grep -iE 'permitroot|passwordauth'`
show the active sshd settings.

`grep -rIn "AKIA" . 2>/dev/null`
hunt for AWS access keys in code.

`gitleaks detect`
scan git history for secrets.

`trivy fs .`
scan a folder for vulnerabilities.

`nmap --script ssl-enum-ciphers -p 443 host`
list a server's TLS ciphers.

`sudo lastlog | head`
last login of each user.

`sudo journalctl _COMM=sudo --since today`
who used sudo today?

`sudo aureport --auth --summary`
auditd authentication summary.

`openssl passwd -6`
generate a SHA-512 password hash.

`sudo chage -M 90 user`
force a password change every 90 days.

SENPAI SAYS
Only audit systems you own or are authorised to test. Log what you changed.

19 BOOT & RESCUE KIT
Resurrection rituals

`journalctl -b -1 -p err`
errors from the previous (failed) boot.

`systemctl --failed`
units that failed to start.

`findmnt --verify`
check fstab for mistakes before reboot.

`blkid`
UUIDs to compare with /etc/fstab.

`sudo update-grub`
Debian/Ubuntu: regenerate the GRUB config.

`sudo grub2-mkconfig -o /boot/grub2/grub.cfg`
RHEL family: regenerate GRUB config.

`sudo grub-install /dev/sda` DANGER
reinstall the bootloader.

`sudo update-initramfs -u`
Debian: rebuild the initramfs.

`sudo dracut -f`
RHEL: rebuild the initramfs.

`efibootmgr -v`
show UEFI boot entries.

`systemd.unit=rescue.target`
kernel parameter (edit in GRUB): rescue mode.

`rd.break`
RHEL: stop in initramfs to reset root password.

`mount -o remount,rw /sysroot`
make root writable in that shell.

`chroot /sysroot`
enter the installed system.

`touch /.autorelabel`
fix SELinux labels on next boot.

`mount --bind /dev /mnt/dev`
prepare a chroot from a live USB.

`chroot /mnt`
work inside the broken system.

`fsck -f /dev/sdX1` DANGER
force a filesystem check (unmounted!).

SENPAI SAYS
Practise rescue steps in a VM first. Don't learn them during an outage.

20 DOCKER TRICKS
Container cunning

`docker run --rm -it --entrypoint sh image`
override the entrypoint to explore.

`docker run --network host image`
share the host network (Linux).

`docker run --read-only --tmpfs /tmp image`
immutable root filesystem.

`docker run --user 1000:1000 image`
run as a non-root user.

`docker run --cap-drop ALL image`
drop all Linux capabilities.

`docker exec -u root -it c sh`
root shell in a running container.

`docker ps --format 'table {{.Names}}\t{{.Status}}'`
custom columns.

`docker inspect -f '{{.State.Health.Status}}' c`
container health status.

`docker stats --no-stream`
one usage snapshot, no live refresh.

`docker logs --since 10m c`
logs from the last 10 minutes.

`docker diff c`
files changed inside a container.

`docker events`
live stream of Docker events.

`docker build --target build -t app:dev .`
build only one stage.

`docker compose up --build -d`
rebuild and start in background.

`docker compose exec svc sh`
shell into a compose service.

`docker compose run --rm svc cmd`
one-off command in a service.

`docker compose down -v` DANGER
stop the stack AND delete its volumes.

`docker image ls --filter dangling=true`
untagged leftover images.

`docker context ls`
list Docker endpoints (remote hosts).

`docker buildx imagetools inspect image`
view a multi-arch manifest.

SPELL SCROLL | Dockerfile (multi-stage)

```
FROM golang:1.22 AS build
WORKDIR /src
COPY . .
RUN CGO_ENABLED=0 go build -o app

FROM gcr.io/distroless/static
COPY --from=build /src/app /app
USER 65532
ENTRYPOINT ["/app"]
```

SENPAI SAYS
Prefer multi-stage builds: compile in a fat image, ship in a tiny one.

21 KUBECTL OUTPUT MAGIC

Bending the output

`kubectl get pods -w`
watch changes live.

`kubectl get pods --sort-by=.status.startTime`
sort by start time.

`kubectl get pods -o custom-columns=NAME:.metadata.name,NODE:.spec.nodeName`
choose your own columns.

`kubectl get pods -o jsonpath='{.items[*].metadata.name}'`
extract values with JSONPath.

`kubectl get pods -o name`
just resource names, for scripting.

`kubectl get pods --no-headers | grep -v Running`
pods that are not Running.

`kubectl get nodes -o wide`
nodes with IPs, OS and runtime.

`kubectl get all -n ns`
pods, services, deployments in a namespace.

`kubectl get deploy -A -o wide`
deployments across all namespaces.

`kubectl get events -n ns --field-selector type=Warning`
only warning events.

`kubectl api-resources --namespaced=true`
what resource types exist?

`kubectl wait --for=condition=Ready pod/p --timeout=60s`
block until a pod is ready.

`kubectl logs -l app=web --tail=20 --prefix`
last lines from every matching pod.

`kubectl top pod --containers`
usage per container.

`kubectl get crd`
list custom resource definitions.

`kubectl explain deploy.spec.strategy`
built-in docs for a field.

`source <(kubectl completion bash)`
enable tab completion.

SENPAI SAYS
JSONPath and custom-columns turn kubectl into a tiny query language.

22 KUBERNETES OPS TRICKS

Fleet wizardry

`kubectl create deploy web --image=nginx --dry-run=client -o yaml > d.yaml`
scaffold a manifest.

`kubectl port-forward svc/web 8080:80`
reach a Service from localhost.

`kubectl exec deploy/web -- env`
run a command in a deployment's pod.

`kubectl scale deploy web --replicas=0`
pause an app without deleting it.

`kubectl patch deploy web -p '{"spec":{"replicas":2}}'`
small change without editing YAML.

`kubectl label pod p env=dev`
add or change a label.

`kubectl annotate pod p note=test`
add an annotation.

`kubectl create job t --from=cronjob/name`
run a CronJob once, now.

`kubectl edit deploy web`
edit live in your editor.

`kubectl rollout pause deploy/web`
pause a rollout to batch changes.

`kubectl rollout resume deploy/web`
continue a paused rollout.

`kubectl debug node/n -it --image=busybox`
debug a node via a debug pod.

`kubectl get --raw '/readyz?verbose'`
API server readiness checks.

`kubectl auth can-i --list`
everything you are allowed to do.

`kubectl create secret generic s --from-literal=k=v`
quick secret.

`kubectl create configmap c --from-file=app.conf`
ConfigMap from a file.

`kubectl config view --minify`
current context's config only.

`kubectl delete pod --field-selector=status.phase==Failed` DANGER
clean up failed pods.

`kubectl replace --force -f f.yaml` DANGER
delete and recreate from a file.

SENPAI SAYS
Scaffold YAML with --dry-run=client -o yaml instead of writing it from scratch.

23 GIT TRICKS II

More time magic

`git log --graph --oneline --all`
branch graph of the whole repo.

`git log --author="name" --since="2 weeks ago"`
filter by author and time.

`git log -S"text"`
commits that added or removed text.

`git log --stat`
files changed per commit.

`git shortlog -sn`
contributors ranked by commits.

`git diff --name-only main`
just the changed filenames.

`git branch --merged | grep -v main | xargs git branch -d`
clean up merged branches.

`git switch -`
go back to the previous branch.

`git add -p`
stage changes chunk by chunk.

`git reset --soft HEAD~1`
undo last commit, keep changes staged.

`git restore --staged file`
unstage a file.

`git stash -u`
stash including untracked files.

`git commit --fixup sha`
mark a commit as a fix for another.

`git rebase -i --autosquash main`
squash fixups automatically.

`git grep "text"`
fast search in tracked files.

`git blame -L 10,20 file`
blame only lines 10 to 20.

`git bisect run ./test.sh`
automate bisect with a test script.

`git describe --tags`
nearest tag plus commit count.

`git rev-parse --short HEAD`
short hash, handy for versions.

`git archive --format=zip HEAD > src.zip`
zip the current commit.

`git config --global pull.rebase true`
rebase instead of merge on pull.

`git remote set-url origin URL`
change the remote address.

`git push origin --delete branch` DANGER
delete a remote branch.

SENPAI SAYS
git reflog plus git reset can rescue almost any mistake. Learn both.

24 TERRAFORM PRO TRICKS

IaC, level two

`terraform plan -target=addr`
plan just one resource.

`terraform apply -var="env=dev"`
pass a variable inline.

`terraform apply -var-file=prod.tfvars`
use a variable file.

`terraform output -json | jq`
outputs as JSON for scripts.

`terraform show -json tfplan | jq`
inspect a saved plan as JSON.

`terraform plan -destroy`
preview what destroy would remove.

`terraform state mv a b`
rename a resource without recreating.

`terraform state rm addr` DANGER
forget a resource, keep it in the cloud.

`terraform graph | dot -Tpng > g.png`
draw the dependency graph.

`terraform fmt -check`
fail CI if code isn't formatted.

`terraform providers lock`
write the provider lock file.

`terraform init -reconfigure`
reset backend configuration.

`terraform init -migrate-state`
move state to a new backend.

`terraform workspace select dev`
switch workspace.

`terraform apply -refresh=false`
skip refresh for a faster apply.

`terraform test`
run built-in tests (1.6+).

`tflint`
catch provider-specific mistakes.

`terraform-docs markdown .`
auto-generate module docs.

`infracost breakdown --path .`
estimate monthly cost.

`TF_VAR_name=value terraform plan`
set a variable from the environment.

SENPAI SAYS
Use -target only for emergencies. Normal work should plan the whole graph.

25 ANSIBLE PRO TRICKS

Automation, level two

`ansible all -m setup -a 'filter=ansible_distribution*'`
facts about each host.

`ansible web -b -m service -a 'name=nginx state=restarted'`
restart a service on many hosts.

```
ansible web -m copy -a 'src=a dest=/tmp/a'
```
push a file to hosts.

```
ansible web -m shell -a 'df -h /'
```
run a shell command on all web hosts.

```
ansible-playbook site.yml --list-hosts
```
who would be targeted?

```
ansible-playbook site.yml --list-tasks
```
what would run?

```
ansible-playbook site.yml --start-at-task "Name"
```
resume from a task.

```
ansible-playbook site.yml --step
```
confirm each task interactively.

```
ansible-playbook site.yml -b -K
```
become root, ask for the sudo password.

```
ansible-playbook site.yml --forks 20
```
more hosts in parallel.

```
ansible-playbook site.yml --syntax-check
```
validate syntax only.

```
ansible-inventory --graph
```
tree of groups and hosts.

```
ansible-config dump --only-changed
```
settings you changed.

```
ansible-galaxy collection install community.general
```
install a collection.

```
ansible-vault view secrets.yml
```
read an encrypted file.

```
ansible-vault rekey secrets.yml
```
change the vault password.

```
molecule test
```
test roles in containers.

```
ansible-pull -U repo.git
```
hosts pull and apply their own config.

SENPAI SAYS
--check --diff first, then run for real. Make that your habit.

26 AWS TRICKS
Sky kingdom: Amazon II

```
aws ec2 describe-instances --filters Name=tag:Env,Values=prod --query 'Reservations[].Instances[].[InstanceId,PrivateIpAddress]' --output table
```
prod instances as a table.

```
aws s3 cp s3://bucket/key - | head
```
stream an object to stdout.

```
aws s3 ls s3://bucket --recursive --summarize | tail -2
```
total objects and size.

```
aws ssm get-parameter --name /app/db --with-decryption --query Parameter.Value --output text
```
read a secure parameter.

```
aws sts assume-role --role-arn ARN --role-session-name t
```
get temporary credentials.

```
aws ec2 create-snapshot --volume-id vol-123 --description pre-change
```
snapshot before risky work.

```
aws logs filter-log-events --log-group-name g --filter-pattern ERROR
```
search CloudWatch logs.

```
aws autoscaling describe-auto-scaling-groups
```
list Auto Scaling groups.

```
aws lambda list-functions --query 'Functions[].FunctionName'
```
list Lambda names.

```
aws eks list-clusters
```
list EKS clusters.

```
aws ec2 describe-instance-status --include-all-instances
```
health checks for all instances.

```
aws --profile dev --region ap-south-1 ec2 describe-vpcs
```
one-off profile and region.

```
export AWS_DEFAULT_REGION=ap-south-1
```
default region for this shell.

```
aws iam get-account-summary
```
IAM usage and quota overview.

```
aws sso login --profile dev
```
log in with IAM Identity Center.

```
aws cloudformation describe-stack-events --stack-name s
```
why a stack failed.

```
aws ec2 describe-volumes --filters Name=status,Values=available
```
unattached EBS volumes (cost!).

```
aws ec2 describe-addresses --query 'Addresses[?AssociationId==`null`]'
```
unused Elastic IPs (cost!).

SENPAI SAYS
Tag everything (Env, Owner, CostCenter). Tags make cleanup and cost tracking possible.

27 AZURE & GCP TRICKS
Sky kingdoms II

```
az vm list -d --query "[].{name:name,ip:publicIps}" -o table
```
VM names and public IPs.

```
az vm run-command invoke -g rg -n vm --command-id RunShellScript --scripts "uptime"
```
run a script without SSH.

```
az resource list --tag env=dev -o table
```
find resources by tag.

```
az aks show -g rg -n c --query provisioningState
```
cluster state.

```
az account show --query id -o tsv
```
current subscription ID.

```
az group exists -n rg
```
true or false.

```
az vm list-sizes -l eastus -o table
```
available VM sizes.

`az monitor activity-log list --offset 1d`
last day of activity.

`az ad signed-in-user show`
who am I logged in as?

`az network public-ip list -o table`
list public IPs.

`gcloud compute instances list --filter="status=RUNNING"`
running VMs only.

`gcloud compute instances list --format="table(name,zone,status)"`
choose output columns.

`gcloud config configurations list`
named gcloud configurations.

`gcloud compute ssh vm --tunnel-through-iap`
SSH without a public IP, via IAP.

`gcloud auth list`
logged-in accounts.

`gcloud container clusters list`
list GKE clusters.

`gcloud logging read "resource.type=gce_instance" --limit 5`
recent VM logs.

`gcloud compute firewall-rules list`
list firewall rules.

`gcloud projects get-iam-policy ID`
who has access to a project.

SENPAI SAYS
Stop non-production VMs at night with a schedule. It is the easiest cloud saving.

28 CLOUD-INIT & BOOTSTRAP
First-boot enchantments

`cloud-init status --wait`
block until first-boot setup finishes.

`sudo cat /var/log/cloud-init-output.log`
output of user-data scripts.

`sudo cloud-init analyze show`
timing of each boot stage.

`sudo cloud-init schema --system`
validate your cloud-config.

`sudo cloud-init clean --logs` DANGER
reset so it re-runs on next boot.

`curl -s http://169.254.169.254/latest/user-data`
AWS user-data (IMDSv1; v2 needs a token).

`curl -H Metadata:true 'http://169.254.169.254/metadata/instance?api-version=2021-02-01'`
Azure instance metadata.

`curl -H 'Metadata-Flavor: Google' http://metadata.google.internal/computeMetadata/v1/instance/name`
GCP instance name.

`packer build template.pkr.hcl`
bake a golden machine image.

`packer validate .`
check Packer config.

`sudo journalctl -u cloud-init`
cloud-init service logs.

`ls /var/lib/cloud/instance/`
cloud-init data for this instance.

SPELL SCROLL | cloud-config (user-data)

```
#cloud-config
package_update: true
packages:
  - nginx
  - fail2ban
users:
  - name: deploy
    groups: sudo
    ssh_authorized_keys:
      - ssh-ed25519 AAAA... you@laptop
runcmd:
  - systemctl enable --now nginx
```

SENPAI SAYS
Bake slow, stable setup into an image (Packer). Keep cloud-init for the small stuff.

29 PROMQL & LOGQL
Reading the metrics runes

`up == 0`
targets that are down.

`sum by (job) (up)`
how many targets are up per job.

`rate(http_requests_total[5m])`
requests per second over 5 minutes.

`100 - avg(rate(node_cpu_seconds_total{mode="idle"}[5m])) * 100`
CPU busy percent.

`node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes`
fraction of memory available.

`histogram_quantile(0.95, sum by (le) (rate(http_request_duration_seconds_bucket[5m])))`
p95 latency.

`increase(errors_total[1h])`
how many errors in the last hour.

`topk(5, rate(container_cpu_usage_seconds_total[5m]))`
five busiest containers.

`predict_linear(node_filesystem_avail_bytes[6h], 24*3600) < 0`
disk full within 24 hours?

`curl 'localhost:9090/api/v1/query?query=up'`
run a query from the shell.

`promtool check rules rules.yml`
validate alert rules.

`{app="web"} |= "error"`
LogQL: lines containing error.

`{app="web"} | json | status >= 500`
LogQL: parse JSON, filter 5xx.

`rate({app="web"} |= "error" [5m])`
LogQL: error rate over 5 minutes.

`logcli query '{app="web"}' --limit=20`
query Loki from the CLI.

SPELL SCROLL | rules.yml (alert)

```
groups:
  - name: node
    rules:
      - alert: HighCPU
        expr: 100 - avg(rate(node_cpu_seconds_total{mode="idle"[5m])) * 100 > 80
        for: 10m
        labels: {severity: warning}
```

SENPAI SAYS
Alert on symptoms users feel (errors, latency), not just on CPU numbers.

30 COMBO MOVES II
Boss-battle one-liners

`for h in web1 web2 web3; do ssh $h uptime; done`
run a command on many hosts.

`diff <(ssh a cat /etc/x) <(ssh b cat /etc/x)`
compare a file across two servers.

`while true; do curl -s -o /dev/null -w "%{http_code}\n" url; sleep 1; done`
poor man's uptime monitor.

`watch -n1 'kubectl get pods | grep -v Running'`
watch for unhealthy pods.

`kubectl get pods -A --no-headers | awk '{print $4}' | sort | uniq -c`
pod count by status.

`comm -23 <(sort a.txt) <(sort b.txt)`
lines only in a.txt.

`ls -t | head -5`
the five newest files.

`journalctl -u app -f | grep --line-buffered ERROR`
follow only errors, live.

`curl -s url | jq -r '.[].name' | sort`
list names from an API.

`openssl rand -hex 16 | tee key.txt`
create and save a secret.

`git log --oneline | head -20 | fzf`
pick a commit interactively.

`docker ps -aq --filter status=exited | xargs docker rm` DANGER
remove exited containers.

`grep -rIl TODO . | xargs sed -i 's/TODO/DONE/g'` DANGER
bulk replace across files.

`kill -9 $(pgrep -f stuck-app)` DANGER
force-kill by name match.

SENPAI SAYS
Build one-liners step by step: run the first stage, check, then add the next pipe.

31 ERROR MESSAGE DECODER
Translating the runes

Permission denied (publickey)
Server rejected your key. Check user, key, authorized_keys perms.

Connection refused
Host answered but nothing listens on that port.

Connection timed out
Packets dropped: firewall, security group or route.

No space left on device
Disk full or out of inodes. Run df -h and df -i.

command not found
Not installed or not in PATH. Try which and echo $PATH.

Too many open files
Raise ulimit -n or LimitNOFILE in the unit.

Address already in use
Port is taken. ss -tulpn | grep :PORT.

Read-only file system
Remounted read-only after errors. Check dmesg, fsck.

Cannot allocate memory
RAM and swap exhausted, or overcommit limits.

Temporary failure in name resolution
DNS problem. Check resolv.conf and dig.

Host key verification failed
known_hosts mismatch. Verify, then ssh-keygen -R host.

Segmentation fault
Bad memory access. Check logs, updates, core dumps.

exec format error
Wrong CPU architecture, or script missing a shebang.

bad interpreter: No such file
CRLF line endings or wrong shebang. Use dos2unix.

ImagePullBackOff
Kubernetes can't pull the image: tag typo or registry auth.

CrashLoopBackOff
Container keeps exiting. kubectl logs --previous.

OOMKilled
Container exceeded its memory limit.

Could not get lock /var/lib/dpkg/lock
Another apt is running. Wait; never delete the lock blindly.

fatal: not a git repository
Run inside a repo folder, or git init.

SENPAI SAYS
Read the whole message. The last line is often the real cause.

32 LS -L DECODER
Reading the runes of ls

-rwxr-xr-x
regular file; owner rwx, group r-x, others r-x.

drwxr-x---
directory; owner full, group read+enter, others none.

lrwxrwxrwx
symbolic link (permissions of the target apply).

-rw-r--r--
typical text file: 644.

-rw-------
private file such as an SSH key: 600.

-rwsr-xr-x
setuid bit (s in owner x slot): runs as the owner. 4755.

drwxrwsr-x
setgid directory: new files inherit the group. 2775.

drwxrwxrwt
sticky bit (t): shared dir like /tmp. 1777.

c / b / s / p
character device, block device, socket, named pipe.

r=4 w=2 x=1
add them up: rwx=7, rw-=6, r-x=5, r--=4.

SENPAI SAYS
First character is the file type. Then three triples: owner, group, others.

33 DIRECTORY MAP
Where everything lives

/etc
system-wide configuration files.

/var/log
log files.

/var/lib
persistent app data: databases, Docker.

/usr/bin
most user commands.

/usr/local
software you install yourself.

/opt
optional third-party software.

/srv
data served by this machine.

/home
user home folders.

/root
home of the root user.

/tmp
temporary files, cleared on reboot.

/var/tmp
temporary files kept across reboots.

/boot
kernel, initramfs and bootloader.

/dev
device files: disks, terminals.

/proc
live kernel and process info.

/sys
kernel devices and settings.

/run
runtime data since boot: PIDs, sockets.

/mnt /media
manual / automatic mount points.

/lib /lib64
shared libraries.

SENPAI SAYS
Config goes in /etc, variable data in /var, your own installs in /usr/local or /opt.

34 CONFIG FILES TO KNOW
The sacred texts

/etc/hosts
static hostname to IP mappings.

/etc/fstab
filesystems mounted at boot.

/etc/passwd
user accounts (no passwords).

/etc/shadow
password hashes, root-only.

/etc/group
group definitions.

/etc/sudoers /etc/sudoers.d/
who can use sudo.

/etc/ssh/sshd_config
SSH server settings.

~/.ssh/authorized_keys
public keys allowed to log in.

~/.ssh/config
per-host SSH client options.

/etc/resolv.conf
DNS servers and search domains.

/etc/nsswitch.conf
lookup order: files, DNS, LDAP.

/etc/crontab /etc/cron.d/
system cron jobs.

/etc/environment
system-wide environment variables.

/etc/security/limits.conf
per-user resource limits.

/etc/sysctl.d/
kernel tuning, persistent.

/etc/logrotate.d/
log rotation rules per app.

/etc/systemd/system/
your own unit files and overrides.

~/.bashrc
your shell aliases and settings.

SENPAI SAYS
Back up a config before editing: cp file file.bak. Validate before reloading.

35 LOG FILES TO KNOW

The chronicles

/var/log/syslog
general system log (Debian/Ubuntu).

/var/log/messages
general system log (RHEL family).

/var/log/auth.log
logins, sudo, SSH (Debian).

/var/log/secure
logins, sudo, SSH (RHEL).

/var/log/kern.log
kernel messages.

/var/log/dmesg
boot-time kernel messages.

/var/log/cron
cron activity (RHEL).

/var/log/nginx/
access.log and error.log.

/var/log/apache2/ /var/log/httpd/
Apache logs (Debian / RHEL).

/var/log/apt/history.log
what apt installed or removed.

/var/log/dpkg.log
low-level package actions.

/var/log/journal/
persistent systemd journal.

/var/log/cloud-init-output.log
first-boot script output.

/var/log/wtmp /var/log/btmp
successful / failed logins (last, lastb).

SENPAI SAYS
Debian logs to syslog and auth.log. RHEL logs to messages and secure.

36 MISSION BOARD

Practice quests

Mission 1: Disk detective
Fill a test folder, then find the 5 biggest files with find, du and sort.

Mission 2: Log detective
Generate fake access logs and count status codes with awk and uniq.

Mission 3: Permission puzzle
Create a team group and a shared folder with setgid and ACLs.

Mission 4: Port hunt
Start python3 -m http.server, find it with ss and lsof, curl it, then stop it.

Mission 5: Backup hero
Write a script: tar plus date stamp, run it from cron, rotate old copies.

Mission 6: Service builder
Make a systemd unit for a script. Kill it and watch it restart.

Mission 7: Docker dojo
Containerise a tiny app, run it as non-root with memory limits.

Mission 8: Kubernetes kickoff
Create a kind or minikube cluster. Deploy, expose, scale, update, roll back.

Mission 9: Terraform trial
Create and destroy one small cloud resource. Read the plan carefully.

Mission 10: Break and fix
Fill the disk, stop a service, break DNS. Then fix each using only the shell.

Mission 11: Alert builder
Run Prometheus and node_exporter. Write one alert for disk space.

Boss mission
Deploy one app end to end: Git, CI, Docker, Terraform, Kubernetes.

SENPAI SAYS
Do one mission a week. Write what you learned. Teach it to someone.

